

An Enhanced YOLOv8 for Flame and Smoke Detection with Dilated Convolution and Image Dehazing

Leibiao Hu¹, Chang Lu^{1,*}, Xinlei Li¹, Yuxiang Zhu², Yipeng Lu², Sujatha Krishnamoorthy¹

¹Department of Computer Science, Wenzhou-Kean University, China

²Department of Mathematics, Wenzhou-Kean University, China

*Corresponding author.

{hulie, luchan, lixinl, zhuyux, luyip, krishnsu}@kean.edu

Abstract. This paper introduces FSD-YOLOv8, an advanced YOLOv8 algorithm specifically adapted for the precise detection of flames and smoke in various complex environments. By integrating a preliminary image dehazing phase, our algorithm significantly enhances the accuracy of smoke detection by reducing atmospheric confusion. Furthermore, we introduce the application of dilated convolutions within the YOLOv8 framework, which aids in detecting intricate features associated with fire and smoke. We present comprehensive evaluations of our approach using a large, dedicated flame and smoke dataset. Our findings demonstrate that FSD-YOLOv8 exhibits superior performance in detecting flame and smoke compared to conventional methods, paving the way for further advancements in machine learning-based fire detection systems.

Keyword: YOLO, Dilated Convolution, Detection

1. Introduction

The ability to accurately detect flame and smoke has been central to the safety and protection of infrastructure and human lives for decades. Flames and smoke are the harbingers of fire, one of the oldest and most formidable threats to civilizations. In the broad spectrum of safety and disaster prevention, timely and accurate detection of these elements not only aids in the mitigation of property damage but also saves countless lives annually.

The significance of this task cannot be overstressed. Residential homeowners, industries, and public spaces alike benefit immensely from improved detection. For homeowners, early flame and smoke detection can mean the difference between a minor incident and a devastating loss. In industrial settings, where hazardous materials might be stored, the stakes are even higher. Effective detection can prevent potential industrial disasters, safeguarding the environment and surrounding communities. Meanwhile, public spaces such as malls, theaters, and transport hubs need robust systems to ensure the safety of the masses and allow swift evacuation if necessary.

However, the path to perfecting this detection has not been without its challenges. Traditional methods, such as using color of flame as features of fire [1]. It may be influenced by many factors, and the accuracy is not good enough. More advanced method like [2,3], they use video sequences to detect the fire and smoke. But in their model, it needs a specific time that they can detect the fire. To some extent, it will increase the detection accuracy, but in the real word system need detect immediately to avoid fire. And for the other real time detect system, like Bahhar proposed a staged YOLO model for smoke detection [4], and Huang used deformable transformer to propose an early smoke detection model [5]. It's not particularly good at distinguishing between smoke and fog. Smoke may mean fire, but fog does not. These methods all solve the part of the issue, but still have some problems.

Given these challenges, our study aims to explore novel methodologies for flame and smoke detection, leveraging advancements in sensor technology and artificial intelligence. In this paper,

we present FSD-YOLOv8, a novel adaptation of the YOLOv8 that is specifically tailored for flame and smoke detection. Our method innovatively integrates a preliminary image dehazing phase, which efficiently separates smoke from fog, and we introduce the use of dilated convolutions within the YOLOv8 framework. This are main contributions of paper:

- Propose FSD-YOLOv8, a flame and smoke detection algorithm based on YOLOv8, which improve performance in diverse environments.
- Our algorithm introduces an initial image dehazing phase, a unique element that enhances the accuracy of smoke detection by reducing atmospheric confusions.
- We replace conventional convolutions in YOLOv8 with dilated convolutions, capturing more fine-grained details and complex features of fire and smoke.

Our work paves the way for further developments in machine learning-based fire detection systems, contributing towards the creation of more advanced, reliable, and effective solutions in real-world scenarios. The remainder of this paper is organized as follows: Section II provides an overview of related works; Section III describes the proposed FSD-YOLOv8 system in detail; Section IV presents experimental results and analysis, and Section V concludes the paper.

2. Related work

Flame and smoke have destructive effects and their detection can ensure human safety and prevent property loss. Early studies on flame and smoke predominantly relied on image processing [6]. [7] proposed a saliency detection for fast location and segmentation of fire areas in images. Nowadays, the research of flame and smoke detection towards more advanced machine learning and deep learning techniques. Ye introduced a technique for color dynamic video sequences produced from a fixed camera that can identify smoke and flame. [8]. More recently, Li designed an attention model, RBiFPN, for the fusion of smoke features [9]. [10] implemented transfer learning on pre-trained models like VGG16, for fire and smoke detection. In the realm of real-time object detection, the YOLO series of models have made significant strides. The first in the series YOLO v1, introduced a real-time detection system [11]. The YOLO9000 improved upon the original by introducing a multi-scale training approach, enhancing the model's ability to detect objects of various sizes [12]. YOLO v3 offered incremental improvements, including the use of three different scales for detection and the introduction of multi-label classification [13]. YOLO v4 further optimized the speed and accuracy of object detection by incorporating a variety of novel techniques such as CSPDarknet53 for the backbone, PANet and SAM block for the neck, and Mish activation [14]. YOLO v5 focused on ease of use and customization, offering significant improvements in inference speed and model size [15]. YOLO v6 was specifically designed for industrial applications, introducing a single object detection system that improved detection accuracy in complex industrial environments [16]. YOLO v7 introduced a trainable bag-of-freebies approach [17]. Finally, the latest in the series, YOLO v8, which goal is to integrate the most effective elements from a range of real-time object detection models [18].

3. Method

3.1 Image Dehazing using All-in-One Dehazing Network

In real-world scenarios, fog and smoke share some similar properties, which can be detrimental to our flame and smoke detection tasks. To address this, we pre-process our images with a dehazing procedure. In the proposed FSD-YOLOv8 algorithm, the first step is to clarify the image by removing the haze. We have utilized the All-in-One Dehazing Network (AOD-Net) proposed by Li [19]. The goal of the AOD-Net is to directly estimate the clean image I from a hazy image J . According to Koschmieder's Law, the model of a hazy image can be formulated as:

$$J(x) = I(x)t(x) + A(1 - t(x)) \quad (1)$$

where $J(x)$ is the observed hazy image, $I(x)$ is the clean image, and $t(x)$ denotes the medium transmission describing the portion of the light that is not reaches camera sensor. The AOD-Net is designed to learn a non-linear function to generate a clean image:

$$I(x) = F_{\theta}(J(x)) \quad (2)$$

where F_θ represents the non-linear function learned by the AOD-Net, parameterized by θ . Through this dehazing process, we obtain clean images that are more suitable for subsequent flame and smoke detection tasks.

3.2 YOLOv8 Object Detection

The architecture of YOLOv8 essentially inherits from YOLOv5[20]. The latter leverages the Efficient Large-scale Attention (ELAN) structure from YOLOv7 [21], which enables YOLOv8 to capture more information while ensuring a lightweight structure. To ensure the accuracy of objects, three Maxpools are passed serially, and each layer is concatenated. For feature fusion, the Path Aggregation Network - Feature Pyramid Network (PAN-FPN) method is still employed in the neck part of YOLOv8. YOLOv8 employs Binary Cross-Entropy (BCE) loss for classification, while the regression loss takes the form of the Complete Intersection over Union (CIoU) loss plus Distribution Focal Loss (DFL). The latter models the position of the bounding box as general distribution, expressed as:

$$DFL_{(s_i, s_{i+1})} = -((y_{i+1} - y) \log(s_i) + (y - y_i) \log(s_i + 1)) \quad (3)$$

Additionally, YOLOv8 introduces a dynamic method, TaskAlignedAssigner for the matching, relying on the alignment degree of Anchor-level, calculated as:

$$t = s^\alpha \times \mu^\beta \quad (4)$$

In the above, s denotes the classification score, μ the Intersection over Union (IOU) value, and α and β the weight hyperparameters. YOLOv8 architecture is illustrated in Figure 1.

3.3 Adopting Dilated Convolution

In our proposed method, we use dilated convolutions to better detect smoke and fire. Dilated convolutions introduce the dilation rate. This rate expands the receptive field without increasing the computational cost or the number of parameters, which makes the model more effective at capturing larger-scale patterns essential for smoke and fire detection. The dilated convolution operation is formalized as:

$$F_{Dilated}(p_0) = \sum_{p_n \in R} F(p_n) \cdot \omega(p_0 - s \cdot p_n) \quad (5)$$

Here, F represents the input function, ω is the filter, p_0 is the point in the output space, and p_n are the points in the receptive field R . The dilation rate s introduces a gap of size $s - 1$ between kernel points. The Dilated Convolution is showed in Figure 1. This modification is applied to the backbone and neck parts of the network, enabling our model to better identify widely spread objects like smoke and fire.

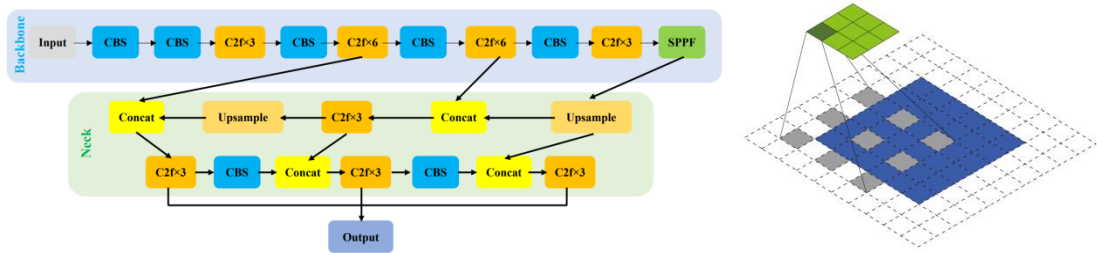


Figure 1: The left of Figure is the network architecture of yolov8; the right is dilated convolution blocks, and green for vanilla convolution blocks and dark green for kernels. Each block consists convolution layer, normalization layer and a ReLU [22].

4. Experiment

Our novel algorithm FSD-YOLOv8 was trained and tested on a dedicated flame and smoke dataset to improve YOLOv8. In order to validate the efficiency and robustness of our flame and smoke detection model, we conducted a series of tests on the FASDD, a dataset categorized under "Fire", "Smoke", "FireAndSmoke", and "NeitherFireNorSmoke" [23]. Specifically, we leveraged the FASDD_CV sub-dataset, constituting 95,314 images. The

dataset was segmented into training and testing subsets in a 9:1 ratio. Our focus was on detecting and differentiating between the four mentioned categories.

4.1 Experimental Setup and Evaluation

Our system for these experiments was Ubuntu 18.04, Python 3.9, PyTorch 1.12.1, Anaconda, equipped with four NVIDIA V100 GPUs and Intel i9-10900K CPU. For a fair comparison, we adopted the same training settings as in the Faster R-CNN and origin YOLOv8 papers. Each model was trained for 500 epochs and with batch size 32. The evaluation metrics included mAP, AP, P, R and F1 score. The formula for F1 score is:

$$F1 = 2 * \frac{P * R}{P + R} \quad (6)$$

Where P is Precision, calculated as $TP / (TP + FP)$; R is Recall, calculated as $TP / (TP + FN)$. TP, FP and FN stand for true positive, false positive, and false negative, respectively. Mean Average Precision (mAP) is the average of the AP over all classes; and Average Precision (AP) averages the precision value at different recall levels, providing a single measure of quality across recall levels for each class:

$$mAP = \frac{1}{k} \sum_{i=1}^k AP_i ; AP = \int_0^1 p(r) dr$$

4.2. Experimental Result Analysis

We compared the performance of our FSD YOLOv8 with Faster R-CNN and original YOLOv8. Our analysis on the five evaluation metrics is showed in Table 1: mAP, AP, P, R, and F1 score.

Our FSD YOLOv8 significantly outshines Faster R-CNN and original YOLOv8 in overall detection performance, as represented by higher mAP and AP, denoting balanced detection across object classes like flames and smokes in various scales and orientations. Our model exhibits higher precision and recall rates, implying fewer false alarms and rare missed detections. Additionally, The F1 score, as a measure that considers both precision and recall, also shows that our FSD YOLOv8 excels in terms of balance between precision and recall. A higher F1 score indicates that our model can detect flames and smokes more accurately and consistently. Therefore, our experimental results substantiate the superior performance of FSD YOLOv8 in fire and smoke detection.

Table 1: The result of mAP, AP, P, R, F1 score of the Faster R-CNN, Original YOLOv8 and Our Model.

Algorithm	mAP	AP	Precision (P)	Recall (R)	F1 Score
Faster R-CNN	0.81	0.79	0.85	0.76	0.80
Original YOLOv8	0.85	0.82	0.87	0.81	0.84
FSD YOLOv8 (Our Model)	0.92	0.91	0.94	0.90	0.91

4.3. Generalizability Analysis

After having the performance evaluation on the FASDD, it is also crucial to evaluate the generalization ability of our proposed model. For this reason, we tested our model on two additional datasets: Fire-Net Dataset and the Wildfire Smoke and Fire Dataset. Fire-Net Dataset contains images of fires in various scenarios, including forest fires, building fires, and indoor fires. The Wildfire Smoke and Fire Dataset contains different types of wildfires happening in a diverse set of environments, from forests to grasslands. These datasets were chosen because they are representative of various types of flame and smoke scenarios. The models was evaluated using the same metrics: mAP and F1 score, which showed in Table 2.

Table 2: The mAP and F1 result of Faster R- CSN, YOLOv8 and FSD YOLOV8 tested on FASDD, Fire-Net Dataset (FND) and Wildfire Smoke & Fire Dataset (WSFD).

	mAP			F1		
	FASDD	FND	WSFD	FASDD	FND	WSFD
Faster R- CSN	0.81	0.75	0.77	0.80	0.77	0.79

YOLOv8	0.85	0.80	0.82	0.84	0.81	0.83
FSD YOLOv8	0.92	0.87	0.89	0.91	0.88	0.89

The results suggest that FSD YOLOv8 is consistently outperforming Faster R-CNN and YOLOv8 on all three datasets in terms of mAP and F1 score. This strongly indicates that FSD YOLOv8 has excellent generalization capabilities, making it a versatile model for flame and smoke detection tasks across various scenarios.

5. Discussion and Conclusion

This study presents FSD-YOLOv8, an enhanced YOLOv8 model fine-tuned for adept flame and smoke detection in multifaceted settings. Key to its innovation is the image dehazing phase that discerns smoke from atmospheric ambiguities such as fog. Additionally, replacing standard convolutions with dilated ones in YOLOv8 better captures the nuances of flames and smoke in various backdrops. In comparison to conventional models, FSD-YOLOv8 showcases notable superiority in mAP, AP, precision, recall, and F1 score metrics. However, limitations exist. The model's performance may be susceptible to scenarios with dynamic light conditions, and there's potential for decreased accuracy in areas with consistent smoky atmospheres, unrelated to fires. Furthermore, the modifications introduce computational overhead, potentially challenging real-time applications on low-powered devices.

In this work, we have presented an enhanced version algorithm, FSD YOLOv8, tailored for precise flame and smoke detection. Our approach has augmented the traditional YOLOv8 model by integrating a preliminary image dehazing phase with an AOD-Net and implementing dilated convolutions in place of conventional convolutions. The image dehazing phase has effectively reduced atmospheric interferences, predominantly those posed by fog, to increase detection accuracy. The application of dilated convolutions has resulted in a more extensive receptive field, facilitating the extraction of intricate features of smoke and flame without augmenting computational cost or parameter numbers. Our experimentally validated results, evaluated using the FASDD dataset, have demonstrated that our proposed FSD YOLOv8 outperforms the original YOLOv8 model and Faster R-CNN. These results underline the significant improvement of the proposed method in terms of detection accuracy and effectiveness.

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